Random Walk

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Introduction

A random walk is a type of stochastic process where a system moves through a sequence of random steps in a given space. It is widely used in finance, physics, and probability theory to model systems that evolve randomly over time.

Definition of a Random Walk

A random walk is a sequence of values S_t where each new value depends on the previous value plus a random change:

$$S_{t+1} = S_t + X_t$$

where:

- S_t is the value at time t.
- X_t is a random variable representing the step taken at time t, often assumed to be independently and identically distributed (i.i.d).

Random Walk Without Drift

In a simple random walk:

$$S_{t+1} = S_t + \epsilon_t$$

where ϵ_t is a mean-zero stochastic term (e.g., normally distributed). This means the expected value remains the same over time:

$$E[S_{t+1} \mid S_t] = S_t$$

Random Walk With Drift

A random walk can have a drift term μ, which represents a systematic upward or downward trend:

$$S_{t+1} = S_t + \mu + \epsilon_t$$

Here, μ shifts the expected future value:

$$E[S_{t+1} \mid S_t] = S_t + \mu$$

If $\mu > 0$, the process has an upward trend; if $\mu < 0$, it trends downward.

Example: Stock Prices and Market Efficiency

In financial markets, stock prices often exhibit random walk behavior, meaning their future values cannot be reliably predicted based on past trends. This concept is closely related to the Efficient Market Hypothesis (EMH), which states that asset prices incorporate all publicly available information.

- **Without drift:** Prices fluctuate unpredictably without any persistent upward or downward trend.
- **With drift:** Over time, prices may show a general upward trend, often observed in equity markets as they grow alongside the economy.

Relationship with Markov and Martingale Processes

Is a Random Walk a Markov Process?

Yes! A Markov process is one where the future state depends only on the present state and not on past values. Since a random walk satisfies:

$$P(S_{t+1} \mid S_t, S_{t-1}, \ldots, S_0) = P(S_{t+1} \mid S_t)$$

it has the Markov property.

Is a Random Walk a Martingale Process?

A martingale is a process where the expected next value equals the current value, given past information:

$$E[S_{t+1} \mid S_t, S_{t-1}, \ldots, S_0] = S_t$$

A random walk **without drift** is a martingale because it satisfies this condition.
A random walk **with drift** is not a martingale since:

$$E[S_{t+1} \mid S_t] = S_t + \mu$$

introducing a predictable component.